

## Unit 5 Guided Notes — The Pythagorean Theorem

$a^2 + b^2 = c^2$ . Add the legs to find the hypotenuse; subtract to find a leg.

🎯 By the end of this unit, I can...

- Use the Pythagorean Theorem to find a missing side of a right triangle
- Apply its converse to test whether a triangle is right
- Solve real-world and coordinate-distance problems

Standards: NC.8.G.6 · NC.8.G.7 · NC.8.G.8

📄 Blank Worksheet (PDF)

✅ Answer Key (PDF)

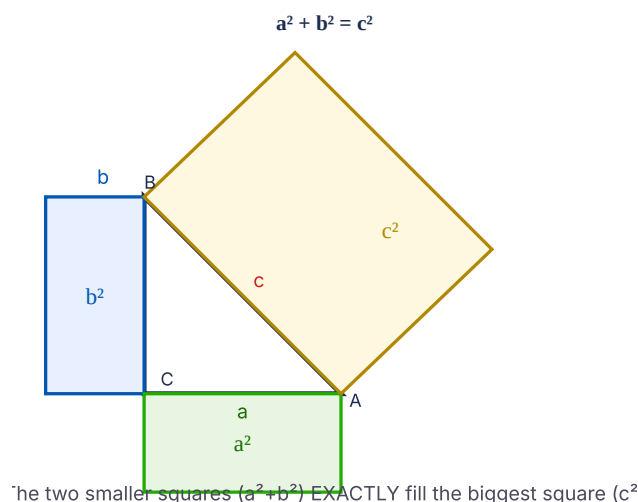
📁 Official PDF Packet

This packet has three parts: **Part 1 — Guided Notes** (fill in the blanks as you learn), **Part 2 — Practice Problems**, and the **Answer Key** built right in.  $a^2 + b^2 = c^2$  — add the legs to find the hypotenuse; subtract to find a leg.

### 1 The Pythagorean Theorem

🔑 **THE THEOREM:** In a RIGHT triangle, the sum of the squares of the two LEGS equals the square of the HYPOTENUSE:  $a^2 + b^2 = c^2$ .

a, b = legs (shorter sides) · c = hypotenuse (longest side, across from the right angle)



⚠️ **IMPORTANT:** The Pythagorean Theorem ONLY works on RIGHT triangles (triangles with a 90° angle)!

Complete each blank on your printed copy, then click to check.

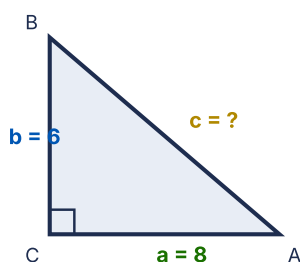
The two shorter sides of a right triangle are called the \_\_\_\_\_ (labeled a and b). The longest side, directly across from the right angle, is the \_\_\_\_\_ (labeled c).

The theorem written as an equation:  $a^2 + b^2 = \underline{\hspace{2cm}}$ . It applies only to \_\_\_\_\_ triangles.

## 2 Finding the Hypotenuse

 **WHEN TO USE:** When you know BOTH LEGS and need to find the HYPOTENUSE. Formula:  
 $c^2 = a^2 + b^2 \rightarrow c = \sqrt{a^2 + b^2}$

### FIND THE HYPOTENUSE



**Example:** Legs are 6 and 8. Find the hypotenuse.

**Step 1 – Write formula:**  $a^2 + b^2 = c^2$

**Step 2 – Substitute:**  $6^2 + 8^2 = c^2$

**Step 3 – Square:**  $36 + 64 = c^2$

**Step 4 – Add:**  $100 = c^2$

✓ **Step 5 – Square root:**  $c = \sqrt{100} = 10$

**Practice — Find the Hypotenuse (c)**

Legs 3 and 4:  $3^2 + 4^2 = c^2 \rightarrow c = ?$

c =

Legs 6 and 8:  $6^2 + 8^2 = c^2 \rightarrow c = ?$

c =

Legs 5 and 12:  $5^2 + 12^2 = c^2 \rightarrow c = ?$

c =

Legs 8 and 15:  $8^2 + 15^2 = c^2 \rightarrow c = ?$

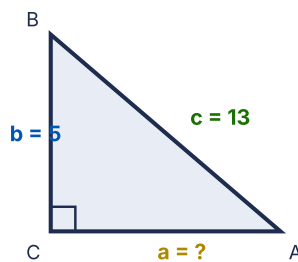
c =

### 3 Finding a Missing Leg

**KEY WHEN TO USE:** When you know ONE LEG and the HYPOTENUSE. Formula:  $a^2 = c^2 - b^2 \rightarrow a = \sqrt{c^2 - b^2}$

**! REMEMBER:** Finding a LEG = SUBTRACT from the hypotenuse squared. Finding the HYPOTENUSE = ADD the two legs squared.

FIND THE MISSING LEG



**Example:** One leg is 5, hypotenuse is 13. Find the other leg.

**Step 1 – Write formula:**  $a^2 + b^2 = c^2$

**Step 2 – Substitute:**  $5^2 + b^2 = 13^2$

**Step 3 – Square:**  $25 + b^2 = 169$

**Step 4 – Subtract 25 (finding a LEG = SUBTRACT):**  $b^2 = 144$

✓ **Step 5 – Square root:**  $b = \sqrt{144} = 12$

**ERROR ANALYSIS** A student finds a missing leg when  $c = 13$ ,  $b = 5$  by writing

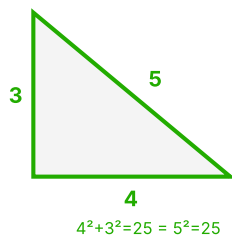
$a^2 = 13^2 + 5^2 = 194$ . What's wrong?

To find a leg you SUBTRACT:  $a^2 = c^2 - b^2 = 169 - 25 = 144 \rightarrow a = 12$ . Adding is only for the hypotenuse.

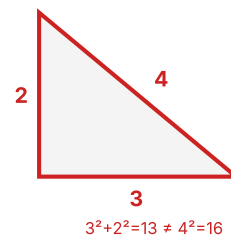
#### 4 Converse of the Pythagorean Theorem

**THE CONVERSE:** Given three side lengths  $a \leq b \leq c$ , test: does  $a^2 + b^2 = c^2$ ? If YES  $\rightarrow$  right triangle. If NO  $\rightarrow$  not a right triangle. Always test the two SMALLER sides against the LARGEST.

3, 4, 5  
RIGHT triangle ✓

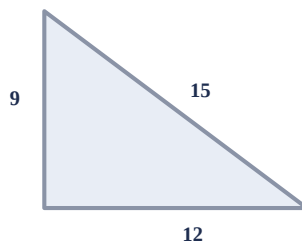


2, 3, 4  
NOT a right triangle ✗



**Example: Sides 9, 12, 15. Is it a right triangle?**

9, 12, 15



**Step 1 – Order the sides:**  $9 \leq 12 \leq 15$ , so the LARGEST is  $c = 15$

**Step 2 – Write the test:** does  $a^2 + b^2 = c^2$ ?

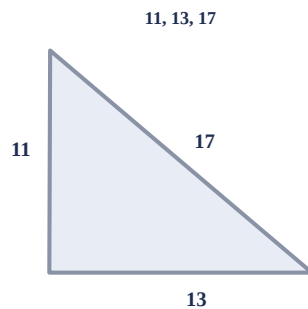
**Step 3 – Substitute the two SMALLER sides:**  $9^2 + 12^2 = ? 15^2$

**Step 4 – Square each one:**  $81 + 144 = ? 225$

**Step 5 – Add the left side:**  $225 = ? 225$

✓ **Step 6 – Compare and conclude:** The two sides are EQUAL, so YES – it is a right triangle.

⚠ **Your eyes cannot do this job.** Every triangle below is drawn to true scale. Look at **11, 13, 17**: it is *not* a right triangle, because  $11^2 + 13^2 = 121 + 169 = \mathbf{290}$  while  $17^2 = \mathbf{289}$ . It misses by one. Drawn accurately it sits two-tenths of a degree off square — nobody alive could spot that. This is exactly why the converse exists: **you must run the arithmetic, every time.**



### Quick Checks

**MC** Sides 5, 12, 13 — right triangle?

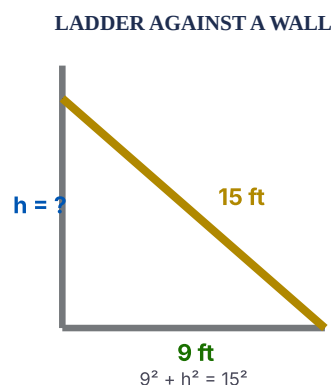
✓  $5^2 + 12^2 = 25 + 144 = 169$ ;  $13^2 = 169$ ;  $169 = 169 \rightarrow$  YES, right triangle.

**MC** Sides 6, 7, 10 — right triangle?

✓  $6^2 + 7^2 = 36 + 49 = 85$ ;  $10^2 = 100$ ;  $85 \neq 100 \rightarrow$  NO, not a right triangle.

## 5 Word Problems

📌 **STRATEGY:** (1) Read carefully (2) Draw & label a diagram (3) Identify legs vs hypotenuse (4) Set up  $a^2 + b^2 = c^2$  (5) Solve step by step.



**Ladder Against a Wall:** A ladder 15 ft long leans against a wall. The base is 9 ft from the wall. How high does it reach?

**Setup:**  $9^2 + h^2 = 15^2$

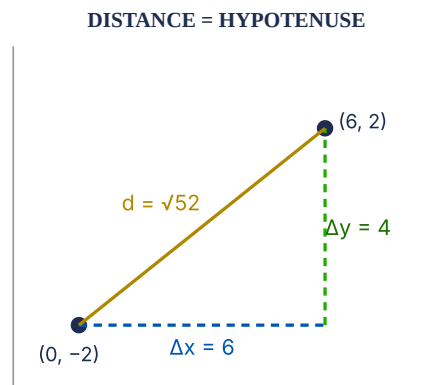
**Square & subtract:**  $81 + h^2 = 225 \rightarrow h^2 = 144$

✓ **Square root:**  $h = \sqrt{144} = 12 \text{ ft}$

## 6 Distance on the Coordinate Plane

 **HOW IT WORKS:** Connect the two points to form a right triangle.  $\Delta x = |x_2 - x_1|$  and  $\Delta y = |y_2 - y_1|$ . Then use  $a^2 + b^2 = c^2$  where  $a = \Delta x$ ,  $b = \Delta y$ , and  $c = \text{distance}$ .

 **NO DISTANCE FORMULA NEEDED!** You already know how to do this — just use the Pythagorean Theorem!



**Example: Distance from (0, -2) to (6, 2)**

**$\Delta x$ :**  $|6 - 0| = 6$

**$\Delta y$ :**  $|2 - (-2)| = 4$

**Setup:**  $\Delta x^2 + \Delta y^2 = d^2 \rightarrow 36 + 16 = 52$

✓ **Answer:**  $d = \sqrt{52} \approx 7.21$

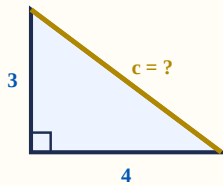
 **WATCH YOUR SIGNS:** When subtracting negative coordinates, subtracting a negative means ADDING! Example:  $5 - (-3) = 5 + 3 = 8$ , NOT  $5 - 3 = 2$ .

## Part 2 — Practice Problems

 Show all work including the formula setup.

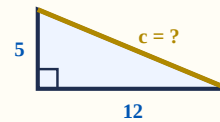
### A. Find the Hypotenuse

1) Legs 3, 4.  $c = ?$



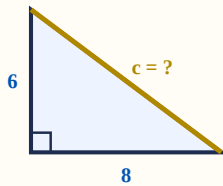
$c =$

2) Legs 5, 12.  $c = ?$



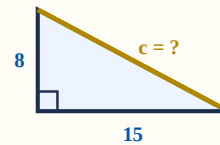
$c =$

3) Legs 6, 8.  $c = ?$



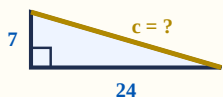
$c =$

4) Legs 8, 15.  $c = ?$



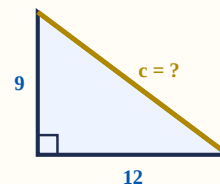
$c =$

5) Legs 7, 24.  $c = ?$



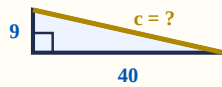
$c =$

6) Legs 9, 12.  $c = ?$



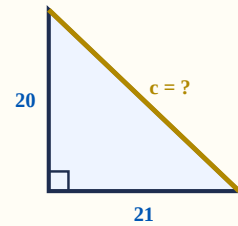
$c =$

7) Legs 9, 40.  $c = ?$



$c =$

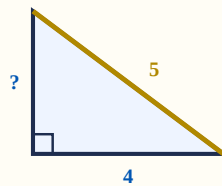
8) Legs 20, 21.  $c = ?$



$c =$

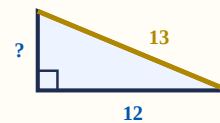
### B. Find the Missing Leg

1) Known leg 4, hyp 5. Leg = ?



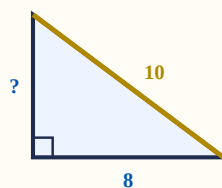
leg =

2) Known leg 12, hyp 13. Leg = ?



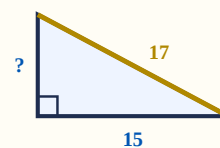
leg =

3) Known leg 8, hyp 10. Leg = ?



leg =

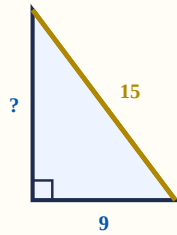
4) Known leg 15, hyp 17. Leg = ?



leg =

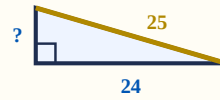


5) Known leg 9, hyp 15. Leg = ?



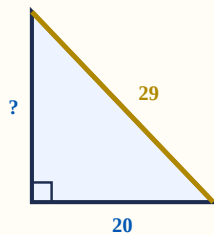
leg =

6) Known leg 24, hyp 25. Leg = ?



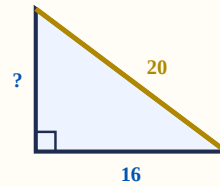
leg =

7) Known leg 20, hyp 29. Leg = ?



leg =

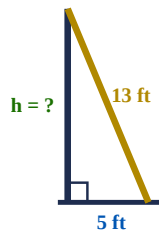
8) Known leg 16, hyp 20. Leg = ?



leg =

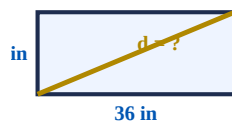
### C. Word Problems

1. A 13-foot ladder leans against a wall. The base is 5 feet from the wall. How high does it reach?



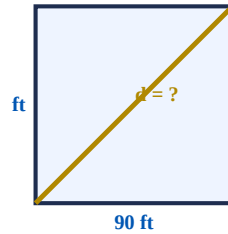
$$h^2 + 5^2 = 13^2 \rightarrow h^2 = 169 - 25 = 144 \rightarrow h = 12 \text{ ft.}$$

2. A rectangular TV is 36 inches wide and 15 inches tall. What is the diagonal length?



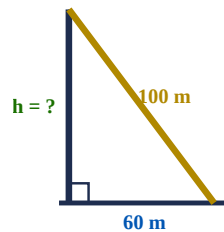
$$d^2 = 36^2 + 15^2 = 1296 + 225 = 1521 \rightarrow d = 39 \text{ in.}$$

**3. A baseball diamond is a square with 90-ft sides. Distance from home to second base?**



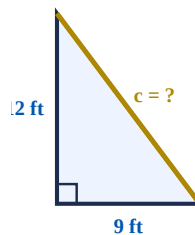
$$d^2 = 90^2 + 90^2 = 16200 \rightarrow d \approx 127.3 \text{ ft.}$$

**4. A kite string is 100 m long. The kite is directly above a point 60 m away. How high is the kite?**



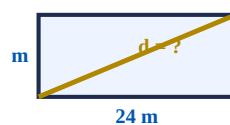
$$h^2 + 60^2 = 100^2 \rightarrow h^2 = 10000 - 3600 = 6400 \rightarrow h = 80 \text{ m.}$$

**5. A support beam covers 9 ft horizontal and 12 ft vertical. How long is the beam?**



$$c^2 = 9^2 + 12^2 = 81 + 144 = 225 \rightarrow c = 15 \text{ ft.}$$

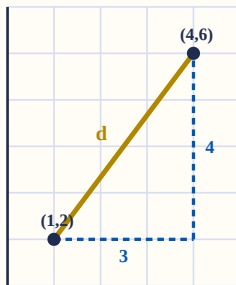
**6. A pool is 24 m long and 10 m wide. A swimmer swims diagonally. How far?**



$$d^2 = 24^2 + 10^2 = 576 + 100 = 676 \rightarrow d = 26 \text{ m.}$$

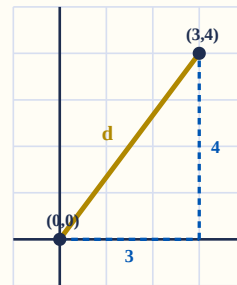
## D. Distance on the Coordinate Plane

1)  $(1,2) \rightarrow (4,6)$ .  $d = ?$



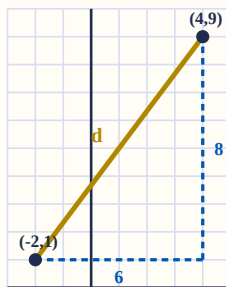
d =

2)  $(0,0) \rightarrow (3,4)$ .  $d = ?$



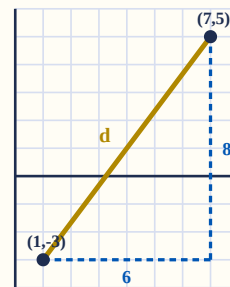
d =

3)  $(-2,1) \rightarrow (4,9)$ .  $d = ?$



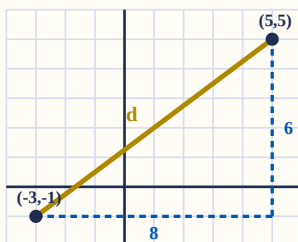
d =

4)  $(1,-3) \rightarrow (7,5)$ .  $d = ?$



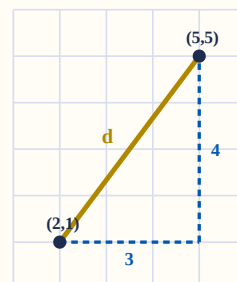
d =

5)  $(-3,-1) \rightarrow (5,5)$ .  $d = ?$



d =

6)  $(2,1) \rightarrow (5,5)$ .  $d = ?$

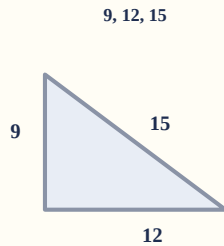


d =

### E. The Converse — Is It a Right Triangle?

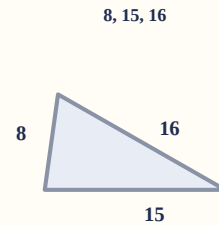
Test the two **smaller** sides against the **largest**. Each figure is drawn to true scale, so do not trust the picture — two of these miss by a hair.

1) Sides 9, 12, 15 — right triangle? (type yes or no)



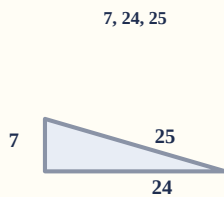
yes / no

2) Sides 8, 15, 16 — right triangle? (type yes or no)



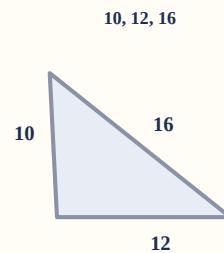
yes / no

3) Sides 7, 24, 25 — right triangle? (type yes or no)



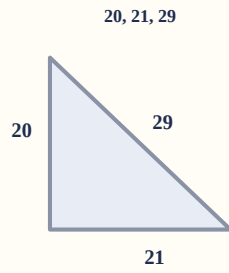
yes / no

4) Sides 10, 12, 16 — right triangle? (type yes or no)



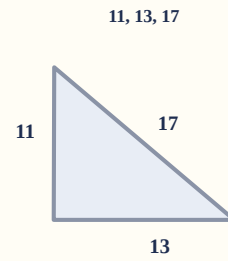
yes / no

5) Sides 20, 21, 29 — right triangle? (type *yes* or *no*)



yes / no

6) Sides 11, 13, 17 — right triangle? (type *yes* or *no*)



yes / no

← Back to Unit 5